
EVIDENCE FOR SIGNIFICANT MULTI-Li⁺ CLUSTERING IN COMMON LITHIUM-ION BATTERY ELECTROLYTES

A PREPRINT

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ABSTRACT

Lithium-ion battery (LIB) electrolytes have been empirically optimized to maximize battery lifetime and charge rate, but a detailed understanding of how their atomic-scale structure controls their electrochemical behaviors remains lacking. Here, we combine machine-learning-enabled atomistic simulations, nuclear magnetic resonance measurements, electrochemical data, and quantum-chemical calculations to reveal previously unstudied multi-Li⁺ complexes without anions that are uniquely facile for reduction. This surprising insight explains the success of empirically-chosen solvent mixtures and provides a foundation for the rational design of improved electrolytes. Furthermore, our machine learning force field (MLFF) based LIB electrolyte simulations accurately reproduce a wide range of condensed phase experiments, despite being parameterized only to quantum chemical gas phase cluster calculations, and hence provide a compelling demonstration that MLFF models developed from first principles can robustly represent a complex chemical system to atomic resolution.

Keywords Lithium-ion · battery · DFT · ab initio · ML · force fields · NMR

1 Introduction

Electrolyte composition plays a crucial role in determining the dynamical and electrochemical properties of modern lithium-ion batteries (LIBs). The bulk solvation structure of the electrolyte is a major determinant of a wide range of properties, including ionic conductivity, viscosity, salt dissociation and subsequent reactivity, the electrochemical stability window (and thus solid electrolyte interphase (SEI) formation), and the composition of the electrochemical double layer. These properties in turn control most aspects of battery functionality.

Elucidation of bulk electrolyte structure at an atomic level of detail has been challenging to accomplish via a purely experimental approach. Consequently, there has been intense interest in carrying out molecular dynamics simulations of the system, which in principle can yield a complete picture of electrolyte structure and dynamics. However, traditional molecular mechanics force fields, such as those used successfully for biomolecular simulation of protein-ligand binding¹⁻³, may not be well suited to describing the complexities of LIB chemistry. Specifically, the physical chemistry of Li⁺ solvation and dynamics in concentrated, polar organic electrolytes is vastly different from that of aqueous biomolecular species. *Ab initio* molecular dynamics (AIMD) approaches are an alternative, but suffer from severe limitations in the system sizes, simulation timescales, and accuracy of the density functional theory (DFT) functional that can be accessed^{4,5}.

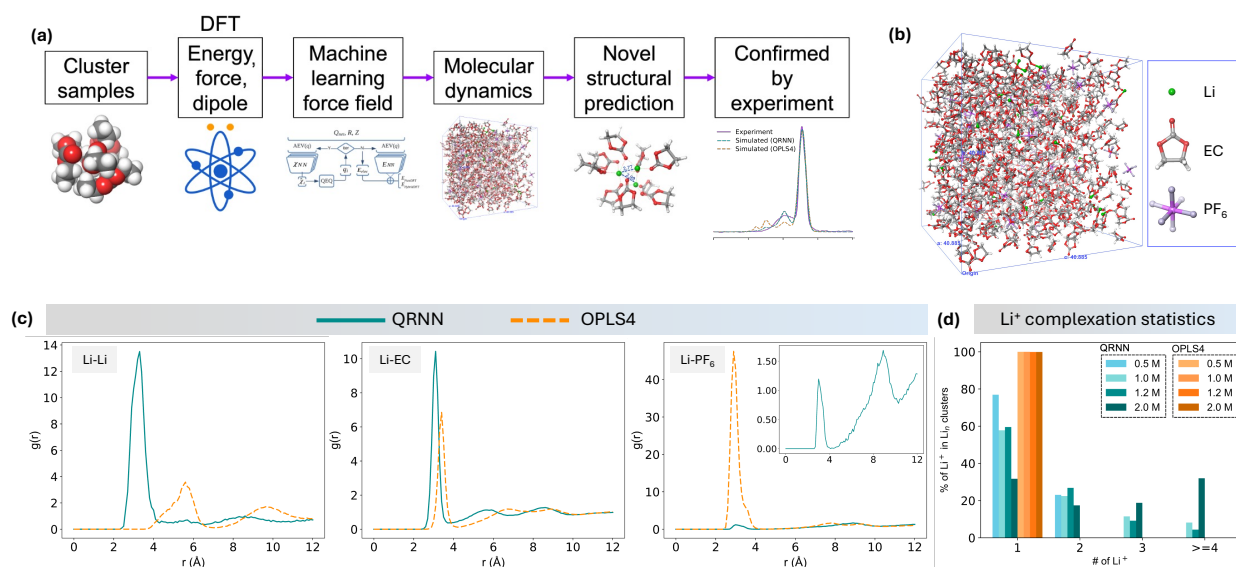


Figure 1: (a) Illustrated workflow for developing the QRNN model, generating computational insights, and corroborating them with experiment. (b) Snapshot of a QRNN MD trajectory. Outset are the constituent ions and molecules. (c) Radial distribution function (RDF) plots of Li⁺ cations with (left to right) other Li⁺ cations, EC solvent molecules, and PF₆⁻ anions calculated using QRNN and OPLS4 production MD trajectories for 1 M LiPF₆ in EC solvent. (d) Statistics of extracted clusters from QRNN and OPLS4 trajectories: relative probabilities of clusters containing various numbers of Li⁺ across different concentrations of LiPF₆.

The past decade has seen the creation and evolution of a new methodology for constructing force fields for complex chemical systems which holds the promise of replicating the accuracy of high level *ab initio* quantum chemistry, while reducing the cost of energy evaluation per configuration of atoms by many orders of magnitude^{6–9}. These methods are based on the use of machine learning force fields (MLFFs) which are parameterized to reproduce quantum chemical relative energies for a large and diverse training set, for example clusters taken from force field simulations^{9–14} or directly from periodic electronic structure calculations^{15–18}. The computational cost of the MLFF is 2–3 orders of magnitude more expensive per energy evaluation than a traditional pairwise additive molecular mechanics force field such as OPLS4²; however, it is still many orders of magnitude less expensive than a purely quantum chemical approach as embodied in AIMD.

In an initial study of LIB electrolytes with a high-dimensional, charge-recursive MLFF, molecular complexes containing multiple Li⁺ ions at short distances (as small as 3 Å) were observed¹¹. In the present work, we report a detailed theoretical analysis of refined MLFF electrolyte model (QRNN), coupled with comparison to NMR experiments, that comprehensively supports the computational discovery of multi-Li⁺ complex species coordinated by ethylene carbonate (EC) solvent molecules. We rigorously demonstrate the agreement of QRNN with high-level quantum chemical calculations and with experimental condensed-phase, structural, and electrochemical properties. Novel insights into the microsolvation structure of Li-ions and their associated redox properties in the battery electrolyte system emerge from the QRNN MD simulations (depicted in Fig. 1(a)). Moreover, improving the accuracy of the MLFF in reproducing reference energetics for these multi-Li⁺ systems resulted in a substantial improvement in predictions for macroscopic electrolyte properties. Our expectation is that simulation technology can now be used to study modified electrolyte compositions and, ultimately, the entire electrochemical system, with the aim of designing LIBs with improved performance characteristics.

2 Li⁺ ion clustering in molecular dynamics

Significant Li⁺ ion clustering was observed in QRNN MD simulations of LiPF₆ in EC at various concentrations. To quantify the liquid electrolyte structure of 1 M LiPF₆ in EC, we computed the radial distribution functions, $g(r)$, of Li⁺ with other Li⁺ ions (Li-Li), with the center of mass of EC (Li-EC), and with the center of mass of PF₆⁻ (Li-P). These plots are shown in Fig. 1(c). The Li-Li RDF from the 1 M QRNN trajectory shows a sharp peak around 3 Å, representative of the presence of multi-Li⁺ clusters in electrolyte solution, which is not seen in the OPLS4

simulations (matching results in the literature for other versions of OPLS¹⁹). The Li-EC RDFs from the QRNN and OPLS trajectories are similar, representing first shell binding of solvent to Li, but the Li-P RDFs diverge considerably, reflecting the separation of the Li⁺-PF₆⁻ ion pairs in the QRNN simulations as noted above.

The statistics pertaining to the molecular complexation of Li⁺ in different concentrations of LiPF₆ in EC, specifically the fractions of Li⁺ in different Li_n clusters are shown in Fig. 1(d). In the QRNN simulations, multi-Li⁺ complexes appear frequently with more than one Li cation in close proximity (< 4.5 Å), with increasing frequency as the concentration increases. This is in contrast to the OPLS4 simulations, in which the clusters overwhelmingly contain one Li⁺ ion (> ~99%); a significant proportion of clusters in the QRNN simulations contain 2-3 Li⁺ ions (and some contain 4-5 Li⁺ ions, although with lower frequency, except at 2 M concentration). At 1.2 M concentration, we observe 40% of the Li⁺ ions to be present in multi-Li⁺ clusters during the QRNN trajectory. This translates to an average of 1.3 Li⁺ ions per Li⁺ complex in solution. Furthermore, the proportion of multi-Li⁺ complexation in EC appears to scale with concentration. At 2 M LiPF₆ more than 68% of the Li⁺ ions are in multi-Li⁺ complexes whereas at 0.5 M that proportion is around 23%.

It is clear from the above data that the solvation structure predicted in our recent QRNN MD is radically different from that observed in other MD simulations, which determines the observed battery properties. The following sections demonstrate the validity of the QRNN results via quantum-chemical calculations, macroscopic electrolyte property predictions, and comparison of experimental and theoretical NMR spectra.

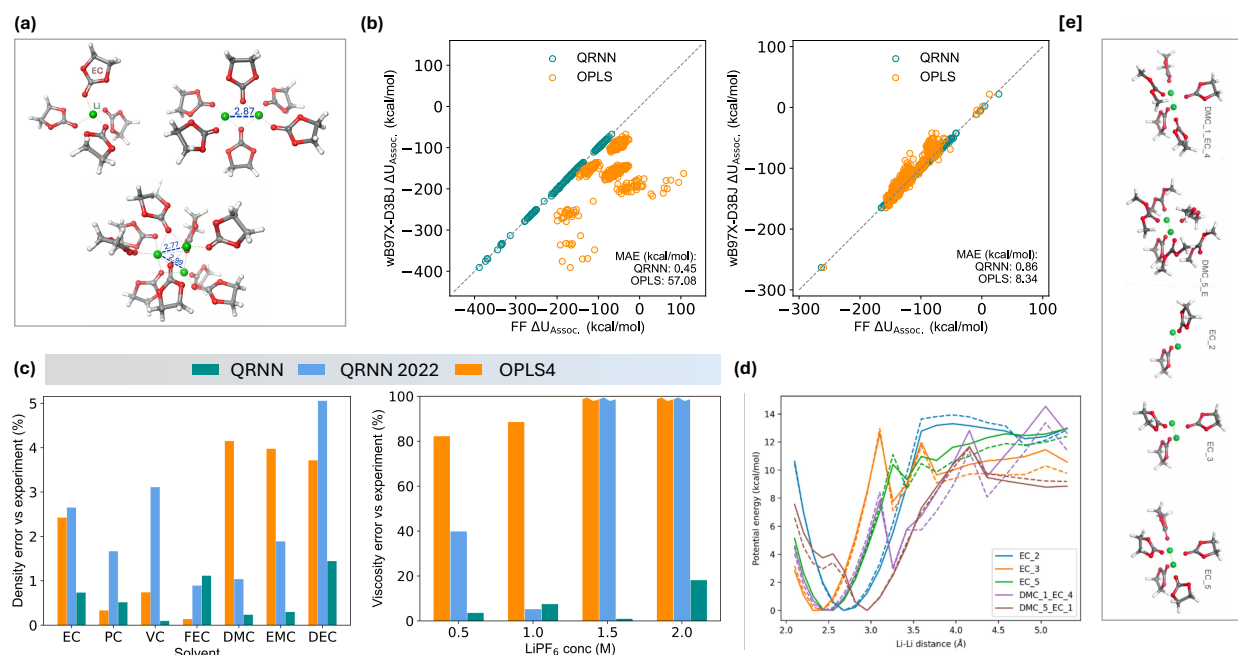


Figure 2: (a) Representative clusters obtained from QRNN simulations, ranging from one to three Li⁺ ions. (b) Parity plots of association energies ($\Delta U_{Assoc.}$) for QRNN (left) and OPLS4 (right) trajectory configurations calculated using DFT (ω B97X-D3BJ/def2-TZVPD) and force field (FF) methods (QRNN and OPLS4). QRNN is more accurate both for its own MD geometries and for the OPLS MD geometries. (c) (Left) Error in predicted density for pure solvents commonly used in electrolyte solutions (lower is better)^{20–22}. (Right) Error in predicted viscosity for LiPF₆ salt dissolved in 1:1 mixed EC:DMC (dimethyl carbonate), with respect to the concentration of dissolved salt (lower is better)^{21–25}. Our previous predictions were accurate only at concentrations at or below 1.0 M; now we have achieved accuracy up to the highest practical salt concentrations. (d) Internal energy versus Li-Li distance scans for QRNN (dashed lines) vs DFT (solid). The jaggedness of the curves is a consequence of the unconstrained solvent molecules adopting different configurations at different Li-Li distances. The plot legend details the carbonate ligands accompanying the 2 Li⁺ ions in the clusters. (e) Illustrations of the molecular clusters for which the Li-Li distance scans were performed.

3 Model validation

3.1 Overview

Below we described three complementary approaches to the validation of our QRNN model for the LIB electrolyte. First, we assess the accuracy with which the MLFF reproduces accurate quantum chemical calculations (DFT which has been calibrated by comparison to DLPNO-CCSD(T) and AFQMC calculations²⁶) for Li-Li distance scans and for the energetics of structures extracted from the QRNN and OPLS4 simulations. These results are contrasted with a similar assessment for OPLS4. In section 3.3, QRNN simulations are benchmarked for a wide range of condensed phase properties in comparison with OPLS4, a 2022 version of QRNN, and experimental data. It should be emphasized that no experimental input was used in constructing the QRNN potential energy surface, making this a fully independent test set. Finally, we demonstrate that the structure of the electrolyte from QRNN yields a ¹³C NMR signal in strong agreement with accompanying experiments across three LiPF₆ concentrations.

3.2 *Ab initio* validation of QRNN for the prediction of Li complexation in solution

First, we selected representative subsets of the Li-centered clusters from both QRNN and OPLS4 trajectories (necessary since the structures are radically different) for further validation against DFT. We calculated the gas-phase electronic energies of association for these subsets of complexes using QRNN, OPLS4, and DFT (ω B97X-D3BJ with def2-TZVPD basis set), the methodology on which the QRNN model was trained. As shown in Fig. 2(b), the association energies calculated with QRNN agree within chemical accuracy (< 1 kcal/mol) with those from DFT for both QRNN- and OPLS4-predicted complexes, despite their very different structures. In contrast, the errors obtained using OPLS4 are 10-100 $\times$ larger, and OPLS4 does particularly poorly at calculating the energies of the multi-lithium clusters. Validation of our DFT methodology to within a few kcal/mol against CCSD(T) and AFQMC calculations is presented in Appendix A.2 and in Debnath *et al.*, respectively²⁶.

We also considered Li⁺-Li⁺ distance scans in the presence of solvent molecules. We created a test set of five such scans, three with pure EC and two with an EC/DMC mixture, by randomly packing solvent molecules and ions. DMC was introduced to test whether minima would persist in the presence of a less polarizable linear carbonate. Each scan is relaxed first with OPLS then with QRNN, and finally evaluated with DFT at the ω B97X-D3BJ/def2-TZVPD level. The results are shown in Fig. 2(d). A clear minimum-energy configuration emerges at a distance between 2.4 and 3.0 Å for each scan, which QRNN correctly reproduces. These minima, and the fact that QRNN correctly reproduces them, provide validation of the existence of low energy Li⁺-Li⁺ complexation by solvent molecules.

3.3 Simulation of condensed phase properties

As in our earlier work¹¹, we can validate our model by simulating a number of standard condensed phase properties and comparing with experiment. Fig. 2(c) shows results for predicted densities of 7 pure solvents and viscosity calculations as a function of LiPF₆ concentration for a 1:1 EC:DMC mixture. A dramatic improvement in accuracy and robustness as compared to both the results of our previous work¹¹, and those obtained from OPLS4 simulations, is apparent. Our results are improved by the addition of more quantum chemical data and better training methods in producing the QRNN model, and also by the use of a faster implementation of QRNN in the Desmond molecular dynamics engine²⁷ which allows the use of larger simulation boxes and longer timescales. Details of the methodological improvements to both the simulation engine and QRNN development effort are given in section 7.1.

3.4 NMR results and discussion

Neat EC and electrolyte samples were flash-frozen in liquid nitrogen to isolate and investigate the microscopic structure of the solvent as a function of LiPF₆ concentration with ¹H-¹³C cross-polarization (CP) ssNMR. In Fig. 3, the neat EC spectrum shows two sharp resonances; one assigned to the carbonyl carbon at 157.6 ppm and the other corresponding to the ether carbon at 68.0 ppm. Upon addition of LiPF₆, we see that the solvent peaks broaden and shoulders appear in both the carbonyl and ether regions, where shoulder resolution from the parent peak is more apparent for the carbonyl line, indicating that Li-ions interact with the C=O group. These shoulders become more pronounced with increasing salt concentration (from 0.5 M to 1.2 M to 2.0 M), pointing to enhanced ionic aggregation, compositionally distinct solvation structures, and reduced solvent availability per cation.

To assign EC-Li⁺ coordination structures detected in ¹³C ssNMR, NMR calculations are carried out by extracting all Li⁺-centered clusters (here called coordinated Li or CL) from the QRNN trajectory, followed by relaxing the cluster geometries with QRNN and computing NMR chemical shifts using PBE/pcSseg-3²⁸⁻³⁰. For all salt concentrations (0.5 M, 1.2 M, and 2.0 M), the predicted NMR spectrum, excluding the free solvent (FS) contribution, is obtained by

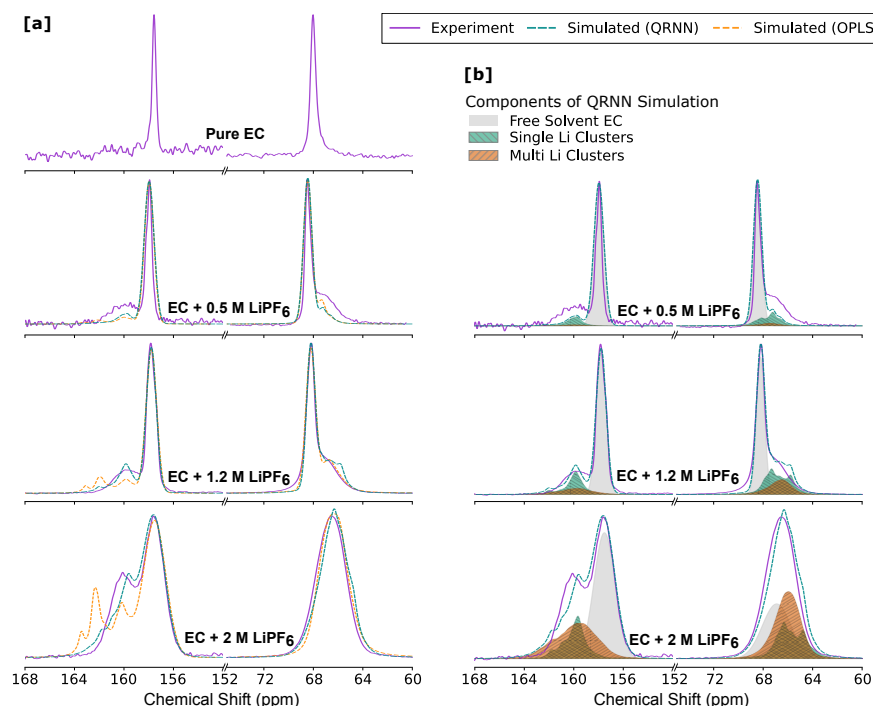


Figure 3: (a) Experimental ¹H-¹³C CPMAS ssNMR (purple lines) for pure EC solvent as well as 0.5 M, 1.2 M, and 2 M LiPF₆ in EC solvent accompanied with NMR spectra simulated with QRNN (dotted green line) and OPLS4 (dotted orange line). Experimental data were recorded at MAS frequency = 20 kHz and T = 116 K after flash freezing to isolate local solvation structures present in EC-based electrolyte solutions. (b) Decomposition of simulated NMR (QRNN) into structural components.

summing individual simulated spectra for all CL clusters in the MD simulation box³¹. The FS contribution is modeled as a single Gaussian line shape. The linewidths of the individual CL peak, and that of the FS peak, are fit to reproduce the experimental data.

To compare the full simulated NMR lineshapes with experiment, it is necessary to place the FS peak relative to the CL peak. For the 0.5 M and 1.2 M concentrations, for both the carbonyl and ether peaks, the relative FS vs. CL peak positions are readily obtained from the experimental spectra. For the 2 M peak, we employ an average of the 0.5 M and 1.2 M shifts (which are very close to each other). We note that rigorous calculation of the position of the FS peak from first principles is not possible because we do not know the structure for the FS region at low temperature. We also present, in Section A.6.3, a hypothesis for a core cluster structure of the pure solvent region, and show that it leads to calculated NMR spectra that are very similar to those utilizing the empirically derived FS shifts presented here. Details of the simulations are discussed further in Section A.6 of the Supplementary Material.

Comparisons between the experimental and simulated spectra from the QRNN trajectory are shown in Fig. 3(a); contributions from different structural components that comprise peak shape, such as free solvent EC, single Li, and multi-Li clusters in the QRNN NMR simulation are shown in Fig. 3(b). The spread in chemical shift of EC molecules in the lithium clusters as well as the gradual growth of the shoulder in carbonyl peak with increasing salt concentration show good agreement with experiments with the QRNN spectra. Moreover, using the average gap obtained from 0.5 M and 1.2 M concentrations yields great agreement of the QRNN spectra with experiment across carbonyl and ether peaks for the 2 M concentration.

When using an identical protocol to generate simulated NMR spectra from the OPLS4 MD simulations where the clusters are relaxed using OPLS4, we see that simulation can adequately predict spectra for samples that predominantly contain single Li⁺-EC structures (0.5 M), but deviate from experiment when multi-Li⁺ clusters are present, especially at carbon sites closest to Li⁺ (Fig. 3(a)). In particular, due to considerably higher irregularity in Li⁺ coordination sites and added representation of PF₆⁻ containing clusters, the OPLS4 simulations fail to reproduce the line shape of the carbonyl peak of the CL clusters when the salt concentration is raised to 1.2 and 2.0 M. For the ether peak, the OPLS4 spectra are qualitatively similar to those obtained from the QRNN simulations. Further insights from simulated NMR spectra are provided in Section A.6 of the Supplementary Materials.

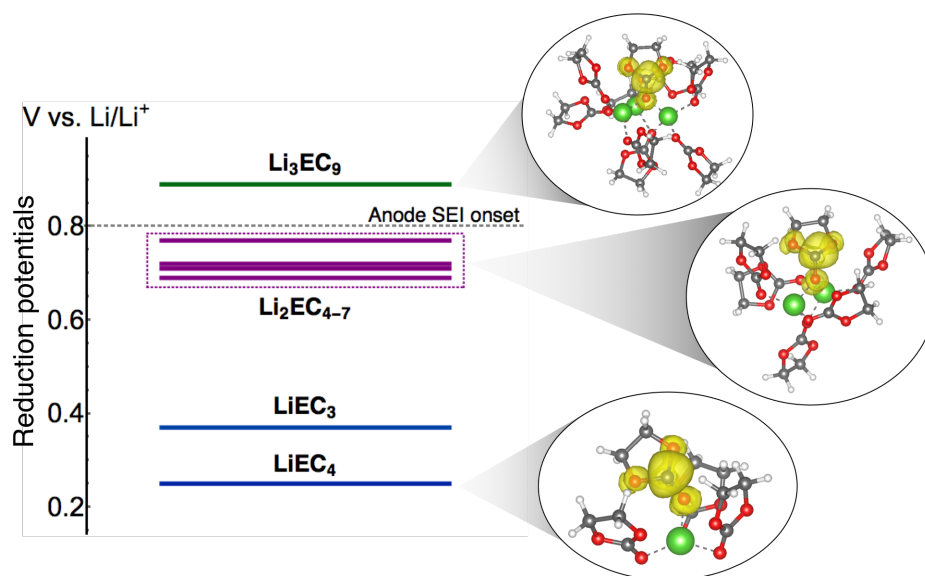


Figure 4: One-electron reduction potentials of several commonly observed Li/EC coordination complexes vs. Li/Li^+ computed with $\omega\text{B97X-D3BJ/def2-TZVPD}$. The dashed line at 0.8 V vs. Li/Li^+ indicates the voltage at which the SEI formation occurs for many common LIB anodes. Outset right are molecular illustrations of select reduced species (LiEC_4 , Li_2EC_6 , Li_3EC_9) that include spin density isosurfaces (yellow) corresponding to the localization of the unpaired, excess electron on each complex. In each case the excess electron is local to the carbonate moiety of one EC ligand. Atom colors: Li (green), C (grey), O (red), H (white). Li-O coordination interactions are depicted as dashed lines.

4 Multi-Li⁺ clusters explain electrolyte reduction potentials

The reduction potentials of species in the LIB electrolyte, which are highly sensitive to their molecular composition and structure, are crucial to the composition and functionality of the resulting anode SEI. We analyzed a series of Li-EC complexes, many of which were highly statistically relevant in the electrolyte from QRNN MD, as described above, to determine their reduction potentials vs. the Li/Li^+ redox couple via quantum chemical calculations. Geometry optimized structures (representing the lowest energy configurations of the oxidized and reduced species) in SMD implicit solvent³², at the $\omega\text{B97X-D3BJ/def2-TZVPD}$ level of theory, are utilized to compute the relevant free energies. Voltages are referenced to Li/Li^+ using a reference $\text{FeCp}^+/_0$ couple. Results for a series of species observed with significant frequency in the simulations are presented in Fig. 4. Further methodological details, benchmarks of the methods for computing referenced reduction potentials, and reduction potentials of additional species are presented in Section A.7 of the Supplementary Material.

From Fig. 4, it is evident that the most common Li_1 complex, LiEC_4 , is reduced at a much lower voltage (less favorably) than the group of $\text{Li}_2\text{EC}_{4-7}$ complexes, the most statistically relevant of which is Li_2EC_7 . Experimentally, SEI formation onset is observed at $\sim 0.8\text{ V}$ vs. Li/Li^+ on both graphite and silicon anodes, according to impedance spectroscopy and electrochemical cycling measurements^{22,33-36}. These data imply that single Li clusters present in the electrolyte would not be spontaneously reduced at this voltage, even allowing for a reasonable statistical distribution of potentials for such complexes. In contrast, the redox potentials for Li_2 clusters fall within 0.1 eV of this value. In addition, the Li_2 complexes appear to demonstrate even more favorable second one-electron reductions (Table 5), adding the possibility that these complexes undergo spontaneous further reduction following the initial one-electron process around 0.8 V vs. Li/Li^+ . Altogether, the more favorable one-electron reduction potentials for multi-Li⁺ complexes suggest that they participate in if not govern the initial anode SEI formation.

The spin density isosurfaces in Fig. 4 demonstrate that the excess electron associated with reduction is localized on the carbonate moiety of an EC ligand, consistent with previous DFT studies of reduced Li_1/EC complexes³⁷. In the reduced multi-Li⁺ complexes, EC ligands that coordinate two or more Li⁺ centers are the electron acceptors, which can help explain the more facile reduction of multi-Li⁺ species. The stabilization of the radical EC^- moiety by multiple coordinating Li⁺ cations likely contributes to a more stable reduced complex.

5 Implications for battery function

Having comprehensively established the presence of multi-Li⁺ clusters in at typical salt concentrations used in LIBs (~ 1 M), we now consider what role these clusters play in the operation of the cell. Upon polarization of the cell, we expect that positively charged species (like those contained in the multi-Li clusters) are attracted to the negatively charged anode surface in the electrochemical double layer and likely contribute to the initial formation of the anode SEI. These species are likely the reactants that reduce at 0.8 V over single Li clusters (between 0.4 - 0.2 V), forming the first layer of the SEI and are responsible for the products reported in literature (LEDC, Li₂CO₃, LBDC, oligomers, etc). We also speculate that the multi-Li⁺ clusters are involved in the transport of Li-ions from cathode to anode and release of these ions at the electrode/electrolyte interface during battery charging. Calculations of the Li⁺ transference number using QRNN are currently in progress.

SEI formation is crucial to proper LIB functioning, and from the electrochemical calculations above, it is reasonable to hypothesize that the reductive decomposition reactions necessary to SEI formation proceed through the multi-Li⁺ cluster species. This deduction in turn provides an explanation as to why EC and related solvents are utilized in the LIB electrolyte. It is almost certainly the case that the high dielectric constant/polarizability of EC (unusual in an aprotic solvent) plays a major role in stabilizing the co-complexation of multiple Li⁺. In essence the EC molecules coordinated to the Li⁺ ions of the cluster provide a potent microsolvation enabling close approach of the positive ions, which would be impossible in the gas phase or in a normal organic solvent. The bridging location of the EC carbonyl oxygen is likely crucial. Further analysis of the decomposition of the multi-Li clusters in the presence of the electrode via QRNN simulation offer the promise of achieving a detailed atomic level picture of the formation of the SEI, and its dependence upon electrolyte composition. This capability in turn opens the door to using large scale GPU-based simulation to carry out electrolyte engineering, thus enabling the design of more efficient and cost effective LIBs and in principle batteries made from other constituents as well.

6 Conclusions

The above analyses of the LiPF₆ in EC electrolyte represent a significant step forward in understanding LIB functionality at an atomic level of detail, providing surprising and previously unknown insights into the microscopic structure of the electrolyte. In future work, we intend to use the QRNN simulation platform to study a wide range of electrolyte solvents, salts, and additives, comparing with experimental data when available. We will furthermore add electrode models, a description of reactive chemistry, and a treatment of the electrode applied voltage, enabling a high degree of realism in the investigation of the early stages of SEI formation to be attained.

In addition to specific implications for understanding the structure and function of the LIB electrolyte, the present work provides proof of concept that quite complex and even surprising phenomena can be accurately modeled via a properly parametrized MLFF without any explicit experimental input. Previous work on water has shown similar capabilities^{38,39}, but the LIB electrolyte system is far more complex in composition and in the forming of unusual covalently bonded species as opposed to the standard non-bonded interactions (van der Waals and electrostatic) that dominate liquid water. The results of this paper suggest that a wide range of important chemical systems, composed of elements from across the full range of the periodic table, can now be effectively addressed via computationally accessible dynamical molecular simulations.

7 Methods

7.1 Machine learning force field and molecular dynamics

Our simulations were performed at 313 K in the NVT ensemble, above room temperature because pure ethylene carbonate is a solid at room temperature. For QRNN, the systems were equilibrated for 1 ns in an NPT ensemble prior to 5 ns production, whereas for the OPLS4 MD we performed an equilibration in NPT for 10 ns followed by 50 ns of production dynamics in NVT. The Li-based molecular clusters were extracted from MD frames using a 4.5 Å radius around Li including molecules with any atoms within this radius.

We carried out molecular dynamics simulations of 1 M LiPF₆ solution in EC solvent using both a machine-learned force field (QRNN) and an empirical force field (OPLS4). We ran NVT trajectories at 313 K for 5 ns using the refined QRNN force field and 50 ns using the OPLS4 force field. Molecular complexes were extracted from these trajectories at evenly spaced time steps based on the immediate coordination environment of the Li-ions in each simulation. The equilibrium dynamics of the same electrolyte yielded vastly different molecular structures depending on the force field used leading to previously unforeseen compositions of lithium complexes.

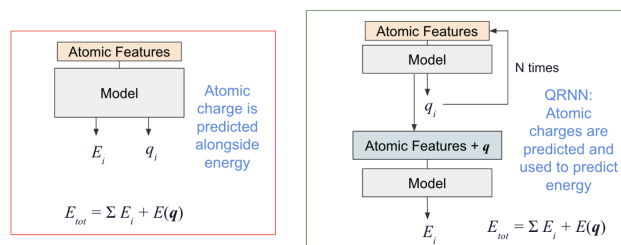


Figure 5: Comparison of a simple charge-predicting MLFF, as compared to the QRNN and its successors which use the predicted charges as input ("charge-aware").

A batch of replicas is launched from the command line onto Google Cloud resources by specifying the desired system composition. Each replica is automatically assigned a unique random seed, which is used to construct an initial simulation cell of approximately 6000 atoms using a Monte Carlo packing algorithm implemented in Schrödinger's Disordered System Builder²⁷. This ensures that each replica has an independent starting point for maximum sampling efficiency. The replicas proceed automatically to an equilibration stage consisting of 1 ns (2M steps) of NPT molecular dynamics with our Desmond QRNN MD engine²⁷.

After equilibration, instead of simply using the final simulation frame for the following steps, our scripts automatically select the trajectory frame with a volume closest to the mean volume over the final 30% of each trajectory. This ensures that the density used for the following step is as close to the true mean density of the model system as possible. After equilibration, the replicas proceed to 5 ns (10 M steps) of NVT molecular dynamics for property calculation.

As in our prior machine learning force field papers^{7,8,11–13}, our model was trained using a fork of the torchANI library^{40,41}, with data labeled at the ω B97X-D3BJ/def2-TZVPD level of theory^{26,42,43} using Psi4⁴⁴ with the same settings as in Stevenson et al¹²:

```
scf_type': 'MEM_DF', 'dft_basis_tolerance': 1e-10,
'ints_tolerance': 1e-10, 'maxiter': 200, 'dft_pruning_scheme': 'robust',
's_orthogonalization': 'partialcholesky', 's_cholesky_tolerance': 1e-6
```

Our prior data from Dajnowicz et al¹¹ was checked using these settings, and inconsistent self-consistent field (SCF) solutions were removed. As in Dajnowicz et al¹¹, we trained the charge model using DFT dipole moments, rather than xTB atomic partial charges⁴⁵. Small-scale geometry optimization and MD for data sampling were performed using the Atomic Simulation Environment library⁴⁶ via the Schrödinger Suite²⁷.

7.2 Sample preparation and NMR experiments

All samples, including 0.5 M, 1.2 M, and 2.0 M LiPF₆ in ethylene carbonate (EC), as well as pure EC solvent, were prepared and packed into 2.5 mm Phoenix zirconia rotors entirely inside an argon-filled glove box to maintain an inert environment and prevent exposure to air and moisture. Samples were flash-frozen prior to insertion into the probe to preserve their room-temperature structure during low-temperature measurements.

All NMR experiments were performed on a Bruker Ascend 600 MHz wide bore compact magnet equipped with a 2.5 mm HFX LT-DNP Phoenix probe, tuned to ¹H/¹³C mode. The magic-angle spinning (MAS) rate was maintained at 20 kHz, and the sample temperature during measurements was estimated to be approximately 116 K. The 90° pulse powers were 172 kHz at 60 W for ¹H and 75 kHz at 60 W for ¹³C. One-dimensional ¹³C spectra were recorded using a standard ¹H-¹³C cross-polarization (CP) experiment with a CP contact time of 3 ms and recycle delay of 300s. Adamantane was used as an external chemical shift reference for ¹³C. The chemical shift of the CH₂ peak (37.75ppm) was adjusted for 116 K using the temperature correction formula reported by Roy Hoffman⁴⁷.

Bulk ¹H T₁ relaxation times were measured using a saturation recovery experiment with CP detection on ¹³C. T₁ values were determined to be 6.3 s for the 2.0 M and 42.2 s for the 1.2 M LiPF₆ samples. For the 0.5 M LiPF₆ and pure EC samples, ¹H T₁ could not be accurately determined as the signal did not saturate even after a 500 s delay.

The differences in ¹H T₁ between the samples may impact the relative intensities of the free solvent peaks and the solvent-ion coordination peaks in ¹H-¹³C CPMAS ssNMR. For example, we see a broad shoulder appear in ¹³C ssNMR upon addition of 0.5 M LiPF₆ to EC, indicating that Li-ions coordinate to the solvent. In principle, the integrated area of this peak indicates the total amount of solvent coordinated to Li. However, the intensity of this shoulder changes very

little upon increasing salt concentration to 1.2 M LiPF₆ where we expect to have additional salt–solvent interactions, likely due to differences in ¹H T₁ and CP transfer efficiency.

7.3 NMR simulation details

All the NMR calculations have been performed using ORCA 6.0⁴⁸. The NMR chemical shielding values are calculated using PBE/pcSseg-3^{28,29}. The isotropic chemical shifts obtained are used to simulate the spectra of different cluster compositions.

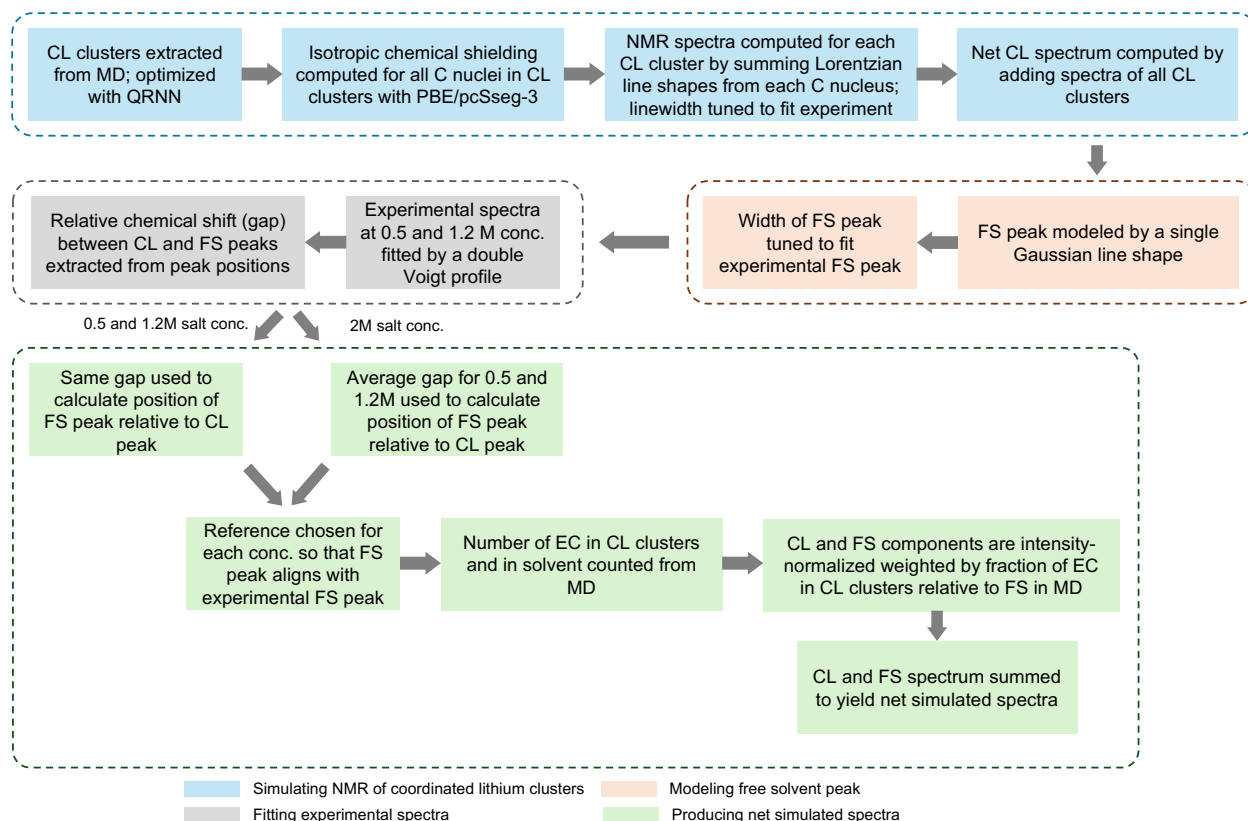


Figure 6: Schematic of the NMR simulation workflow

Two distinct types of species are identified in the MD trajectories, namely coordinated lithium clusters and free solvent molecules. We follow the following steps to simulate the NMR spectra at different salt concentrations using the QRNN or OPLS4 trajectories.

- Coordinated lithium clusters are extracted from the MD trajectories. For the purpose of NMR calculations, the effect of PF₆[−] ions, even when they are not directly part of the Li⁺ complexation environment but are present nearby, are considered. To achieve this, when applicable, one PF₆[−] unit is included beyond the regular cluster when a distance cutoff is satisfied. When there are multiple qualifying PF₆[−] ions, the one closest to the original cluster is included. NMR spectra is computed for each cluster by summing Lorentzian line shape contributions arising from the isotropic chemical shifts of all carbon atoms within the clusters.
- The net contribution of all the coordinated Li clusters in the MD trajectory is obtained by summing the spectra of each individual lithium cluster.
- The number of EC molecules within coordinated lithium clusters is counted along with the number of free solvent EC molecules in the MD trajectory.
- For 0.5 M and 1.2 M salt concentrations, the experimental spectra is fitted by a Voigt profile each for the solvent peak and the coordinated lithium peak⁴⁹. The relative chemical shift among the two peaks is extracted.

- (e) The gap between the peaks is used to calculate the position of free solvent peak with respect to the coordinated lithium peak, calculated earlier, which is modeled by a single gaussian line shape.
- (f) The width of the free solvent peak as well as the individual Lorentzian line shape is tuned to fit the experimental NMR spectra for different peaks at different concentrations.
- (g) The reference is chosen so that the free solvent peak is aligned with the experimental solvent peak. For each concentration, the reference for carbonyl and ether peak is chosen individually⁵⁰.
- (h) The sum of the coordinated lithium peak and the free solvent peak produces the final simulated spectra, to be compared with the experimental NMR spectra. The coordinated lithium and free solvent components in each simulated spectrum are intensity normalized weighted by the fraction of EC molecules in Li clusters relative to those in the free solvent in the MD simulation box.
- (i) For the 2 M concentration, instead of calculating the gap between the coordinated lithium and the free solvent peak from the experimental spectra, we use the average obtained from 0.5 M and 1.2 M and use the same gap relative to the coordinated lithium peak, accounting for the relative positions of the coordinated lithium peaks for 2 M concentration with respect to 1.2 M concentration.

7.4 Reduction potentials versus Li/Li⁺ via ferrocenium/ferrocene

In order to compute reduction potentials of Li-electrolyte complexes on the LIB-relevant reference scale versus Li/Li⁺, we utilize experimental data and computed potentials for the ferrocenium/ferrocene redox couple (denoted hereafter as FeCp^{+/0}) in LIB-relevant electrolytic conditions. We note that this approach is a departure from a significant body of computational literature in which a constant, vacuum-based reference of 1.4 V is applied to the computed redox potentials^{51–54}. Our approach is, however, consistent with emerging best practices of experimental electrochemistry, in which using FeCp^{+/0} is recommended given its versatile solubility and electrochemical reversibility in many different solvents and electrolytes^{54,55}. Given the challenging nature of the transition metal redox chemistry of FeCp^{+/0}, we furthermore impose a correction for the error in the gas phase adiabatic ionization potential of each electronic structure method used⁵⁶.

Thus, the reduction potentials of Li-electrolyte coordination complexes referenced to Li/Li⁺, E_{red} , are computed as follows:

$$\begin{aligned}
 E_{red} &= E_{ref}(\text{FeCp}^{+/0}) + E_{calc}(\text{FeCp}^{+/0}) - E + \delta \\
 E &= -\frac{\Delta G^\circ}{n\mathcal{F}} \\
 \Delta G^\circ &= \Delta G^\circ(gas) + \Delta\Delta E(sol)
 \end{aligned}
 \tag{1}$$

where $E_{ref}(\text{FeCp}^{+/0})$ is the experimental reduction potential of FeCp^{+/0} relative to Li/Li⁺ in the electrolytic conditions of interest, $E_{calc}(\text{FeCp}^{+/0})$ is the calculated, unreferenced reduction potential of FeCp^{+/0}, E is the computed reduction potential of the Li-based complex, and δ is the error in the gas phase adiabatic ionization potential of the electronic structure method. $E_{ref}(\text{FeCp}^{+/0})$ in the present work is taken to be 3.25 V vs. Li/Li⁺ based on cyclic voltammetry of FeCp^{+/0} in 1 M LiPF₆ in 1:1 EC:EMC⁵⁷. In this expression, $E_{calc}(\text{FeCp}^{+/0})$ and E are computed as the solution-phase free energy differences between the reduced and oxidized species. As a consequence of the electrolyte conditions of the reference couple, we employ $\epsilon = 33.6$ for implicit solvation of all species in reduction potential calculations when using both C-PCM and SMD solvation schemes⁵⁸. The additional parameters for the SMD scheme are taken as those previously determined for pure EC⁵⁹. The values of δ for ω B97X and DLPNO-CCSD(T) are taken from previous benchmarking calculations⁵⁶ and are -4.30 kcal/mol and +3.06 kcal/mol, respectively. The reaction free energies associated with each reduction reaction, ΔG° , were computed as the sum of the reaction free energies in gas phase, ΔG° , and the differences in SMD solvation energies between the reduced and oxidized states, $\Delta\Delta E(sol)$. Zero-point energy and entropic contributions to the free energies were obtained by a harmonic frequency calculation of each species at its gas-phase optimized geometry.

The total shift on the computed reduction potentials, $E_{ref}(\text{FeCp}^{+/0}) + E_{calc}(\text{FeCp}^{+/0}) + \delta$, corresponding to the Li/Li⁺ scale are 1.36 V and 1.43 V for DLPNO-CCSD(T)/aug-cc-pVTZ and ω B97X-D3BJ/def2-TZVPD, respectively, using SMD for solvation. While these values are close to the commonly used 1.4 V, they reflect sensitivities to the errors in the underlying methods and moreover could be consistently adopted for any electrolytic conditions so long as the accompanying experimental voltammetry for FeCp^{+/0} vs. Li/Li⁺ has been measured. The reduction potentials in the main text were computed using ORCA 6.0⁴⁸ and the ω B97X-D3BJ/def2-TZVPD level of theory^{42,43}.

8 Data Availability

Cluster geometries extracted from simulation trajectories, used to validate energy predictions against DFT, simulate NMR spectra and compute reduction potentials will be made available upon publication of this manuscript.

9 Code Availability

The MLFF model used to perform molecular dynamics in this work is available in the 25-3 release of Schrödinger Suite under the name QRNN_Electrolyte for molecular dynamics with periodic boundary conditions using Desmond.

10 Acknowledgements

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11 Author Contributions

J.M.S conceptualized the project, developed MLFF methodology, curated data, investigated predictions, validated the model and authored the manuscript. G.A. curated data, developed MLFF methodology, performed MD simulations, analyzed the results and edited the manuscript. L.B. performed NMR experiments, analyzed results and wrote the manuscript. S.D. conceptualized and performed NMR predictions, analyzed the results and edited the manuscript. Z.K.G developed clustering methodology, validated the MLFF model, developed methodology for and performed reduction potential calculations, analyzed the results, authored and edited the manuscript. M.M. developed the clustering methodology, conceptualized and performed NMR simulations, validated the MLFF model, analyzed the results, authored and edited the manuscript. I.Y. performed AIMD simulations analyzed results and edited the manuscript. R.D.G. performed validation of our MLFF and other force fields and edited the manuscript. A.S.A. analyzed results and edited the manuscript. D.S. authored and edited the manuscript. L.M. supervised the project, developed NMR methodology, analyzed results, authored and edited the manuscript. A.U. supervised the project and edited the manuscript. L.D.J. developed MLFF methodology, conceptualized the project and edited the manuscript. M.D.H. supervised the project. R.A.F. conceptualized the project, supervised the project, authored and edited the manuscript and acquired funding.

A Appendix

A.1 Prior work on multi-Li⁺ complexation and Li⁺ solvation structure

There is some literature precedent for multi-Li⁺ complexation, albeit not yet with experimental evidence or in LIB-relevant solvents. Close Li-Li distances in solids have long been known in systems such as lithium carbonate⁶⁰⁻⁶², lithium oxide^{62,63}, and LDA-THF complexes⁶⁴. In liquids, multi-Li⁺ complexes have been suggested by MLFF simulations of concentrated aqueous LiCl⁶⁵ as well as by AIMD simulations of LiTFSI water-in-salt electrolytes⁶⁶. Although both of these studies employ water as the primary solvent, they similarly identify Li-Li distances of less than 3 Å with bridging oxygen atoms, circumstances comparable to our observations for 1 M LiPF₆ in EC. Likewise, the AIMD simulations of bisalt-in-sulfolane electrolytes by Mukherji et al⁶⁷ also showed some multi-Li⁺ complexation, even though the classical force field MD used in the same work excluded these structures.

The separation of Li⁺ ions has been widely assumed in the lithium-ion battery literature, with research focused instead on the other possible components of the solvation shell. Nuclear magnetic resonance (NMR) spectroscopy of Li⁺ ions in solution⁶⁸ has been performed but has been analyzed assuming that the Li⁺ solvation environment does not contain Li⁺, with changes in the solvation shell being assumed to be caused only by CIPs. Han *et al.*⁶⁹ use a combination

of experimental and computational techniques to elucidate the solvation structure, and do consider the possibility of Li⁺ ions being in each other's solvation shells, but only when bridged by PF₆⁻ counterions, forming contact ion pairs (CIPs) that cannot by themselves be the foundation for an effective battery. Hasan *et al.* make further progress by examining Li⁺/Li⁺ distances in classical MD, but their OPLS force field is not capable of producing this type of clustering because of its simple electrostatic repulsion model for Li⁺/Li⁺ interactions¹⁹. Hou *et al.*⁷⁰ and Chen *et al.*⁷¹ assume a consistent single-Li ion environment in the solution and then seek to improve desolvation by solvent engineering. Su *et al.*⁷² likewise assume that the only possible solvent environment for Li⁺ is the solvent, the counterion, and solvent-engineering additives. Chen *et al.*'s review of the literature on non-covalent, non-electrostatic bonding in lithium ions likewise does not cover this effect⁷³.

However, there have also been hints from experiment that an unknown phenomenon in the Li⁺ solvation environment may be present. Avall *et al.*⁷⁴ describe that there are conflicting reports of the coordination number of Li⁺, with some estimates being as low as 2.3 solvent molecules per Li⁺. This finding could be explained by the presence of Li⁺ agglomerations which cause a single solvation shell to be shared between multiple Li⁺ ions. Likewise, Uchida *et al.*⁷⁵ report that calculations of the size of the Li⁺ solvation shell via fluid dynamics (the Stokes radius) and via the expected composition (the van der Waals radius) produce inconsistent results. Once again, the discrepancy could be explained by Li⁺ agglomeration.

Past work assuming total Li⁺ ion separation or only Li⁺ co-complexation in CIPs has been supported by evidence from simulations. It is straightforward to show, as we will, that a traditional fixed-charge force field does not produce Li⁺ cation clusters. However, the degree of Coulombic interaction found in such force fields is unrealistic at close range, causing problems that are typically addressed by fitting smaller atomic partial charges^{76–78}. Only recently have models emerged, such as QRNN^{8,11} and BAMBOO⁹, which attempt to reproduce the nanostructural details of Li⁺ solvation with *ab initio* interatomic potentials, and on long enough timescales, that an interaction such as Li⁺ clustering could be found.

Short Li⁺-Li⁺ distances were recently found for concentrated aqueous electrolytes via *ab initio* molecular dynamics, suggesting that Li⁺ ions can associate at close range in this context⁶⁵. Furthermore, *ab initio*-based machine learning force field simulations of highly concentrated water-in-salts electrolytes demonstrated Li clustering with Li⁺-Li⁺ distances of around 2.5 Å⁷⁹.

A.2 Validation of DFT vs higher-level theory

The choice of DFT functional for both the training of the QRNN MLFF as well as the thermo- and electrochemical analyses of the Li complexes from the electrolyte were based on benchmarking of properties against DLPNO-CCSD(T). Given the discovery of larger and multi-Li complexes in QRNN MD, we expanded the benchmarking of several DFT functionals versus DLPNO-CCSD(T) to the gas phase electronic energies of association to include these clusters, shown in Table 1. The association electronic energies were computed for the geometry-optimized clusters as the association products of the corresponding numbers of Li⁺ cations and EC molecules.

N_{Li}	N_{EC}	DLPNO	ω B97X-D3BJ	B2PLYP+D3	CAM-B3LYP+D3	M062X+D3	PBE0+D3	SCAN+D3	B3LYP+D3
1	1	-51.284	0.55	0.41	1.8	0.45	0.31	1.85	0.64
1	2	-90.0864	0.90	0.93	2.62	1.73	1.32	2.91	0.32
1	3	-116.0119	0.40	2.74	1.49	3.62	2.78	4.29	1.70
1	4	-129.8996	3.57	0.62	4.86	0.49	0.33	0.34	1.13
1	5	-149.4664	4.50	7.40	3.68	7.18	10.06	7.75	8.07
2	4	-157.8703	1.01	6.74	0.42	7.55	8.46	10.78	4.62
2	5	-195.1034	0.57	7.86	0.23	8.72	9.19	11.08	5.88
2	6	-217.8225	2.01	9.46	1.27	11.34	12.72	13.76	7.66
3	4	-101.1481	2.05	5.87	3.07	9.71	8.03	14.48	1.72
3	5	-157.8036	3.34	9.08	1.44	11.26	12.04	16.46	5.37
3	6	-194.6216	0.67	10.18	0.78	13.76	13.47	17.71	6.46
4	6	-97.3587	0.70	9.88	2.10	15.26	13.37	21.14	4.86
MAE			1.69	5.93	1.98	7.59	7.67	10.21	4.04

Table 1: Gas-phase electronic energies of association, ΔU_{assoc} , from DLPNO-CCSD(T)/aug-cc-pVTZ, and the absolute errors, $|\Delta \Delta U_{assoc}|$, from several DFT functionals. The Grimme D3 dispersion correction was used for each DFT method other than ω B97X-D3BJ. For complexes beyond LiEC₄, the ω B97X-D3BJ geometry was used for all other methods. All values are in kcal/mol; the bottom row shows the mean absolute error (MAE) for each functional.

A.3 Parity plots of association energies using alternative DFT functional

In Fig. 7 we show the parity plots of association energies ($\Delta U_{Assoc.}$) for QRNN and OPLS4 trajectory configurations calculated using an alternative DFT functional (CAM-B3LYP-D3BJ⁸⁰/def2-TZVPD) and the same force field (FF) methods (QRNN and OPLS4). Identical to Fig. 2(b), all 983 configurations from the QRNN trajectory and 500 sampled configurations from the OPLS4 trajectory at 1.2 M LiPF₆ concentration in EC are included in the calculation. The association energies calculated are close to ω B97X-D3BJ/def2-TZVPD, producing qualitatively identical trends as before.

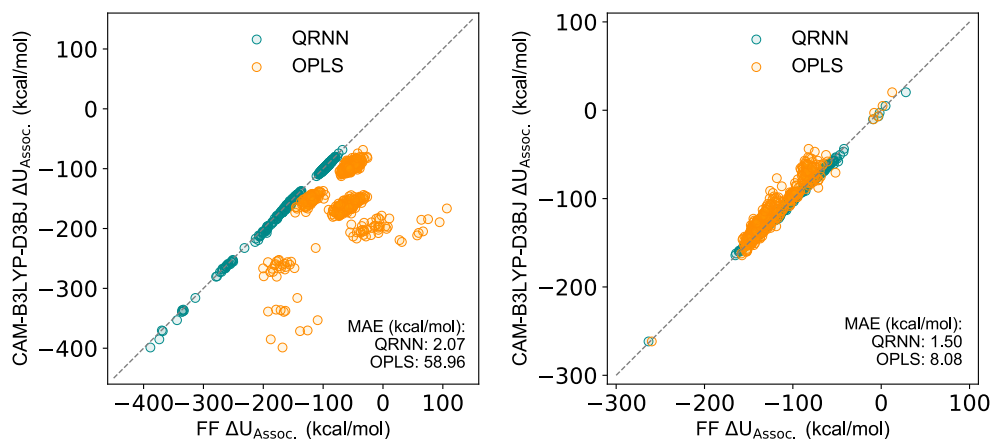


Figure 7: Parity plots of association energies ($\Delta U_{Assoc.}$) for QRNN (left) and OPLS4 (right) trajectory configurations calculated using DFT (CAM-B3LYP-D3BJ/def2-TZVPD) and force field (FF) methods (QRNN and OPLS4). QRNN is more accurate both for its own MD geometries and for the OPLS4 MD geometries with respect to CAM-B3LYP-D3BJ.

A.4 Comparison with AIMD

Due to its high computational costs, ab initio molecular dynamics (AIMD) cannot typically achieve the necessary timescales for equilibrium, condensed-phase aggregation phenomena such as the Li⁺ clustering observed via MLFF-based MD. However, given reasonable starting geometries, we can use AIMD to probe the dynamics of the Li⁺ clusters independently of the assumptions of our model. We used QRNN to create a small solvated cluster of three Li⁺ ions in a box of 154 atoms containing thirteen EC solvent molecules and three PF₆⁻ counterions as shown in Fig. 8. Then we ran NVT AIMD with the PBE functional²⁸, saving trajectory snapshots every 20 fs.

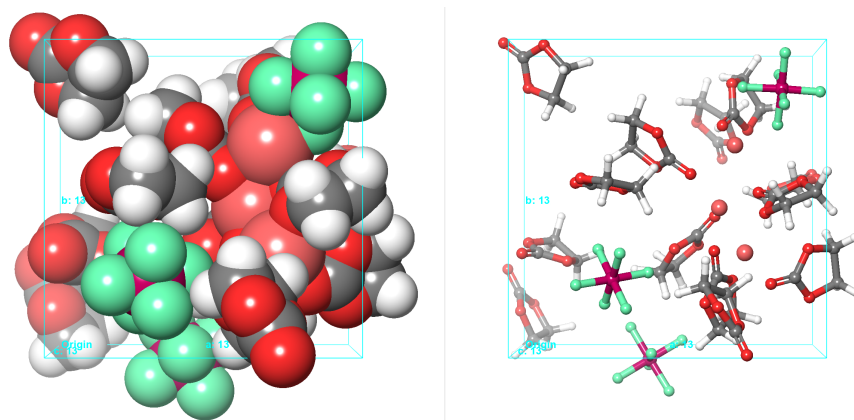


Figure 8: Starting configuration for AIMD: 154 atoms including thirteen EC solvent molecules, three Li⁺ ions, and three PF₆⁻ counterions. The left is a van der Waals radius representation, while the right is the same system in ball-and-stick view.

The resulting Li⁺ distances over time are shown in Fig. 9. Clearly there is no strong repulsion between the Li⁺ ions in this context, and in fact the Li1-Li3 pair shows that the ions are able to re-approach one another after drifting apart. This is despite PBE generally disfavoring Li-Li distances as small as is predicted by QRNN and its underlying ω B97X-D3BJ.

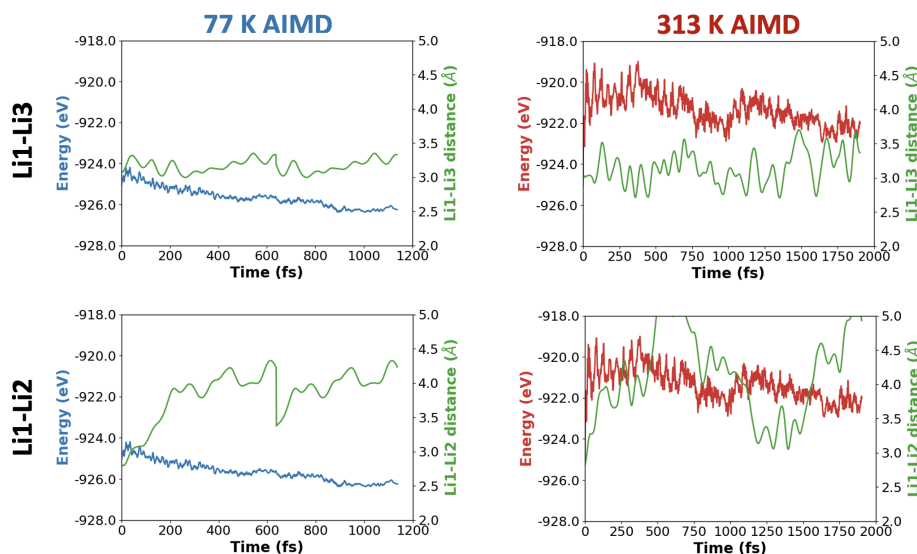


Figure 9: Dynamics of two Li⁺ pairs in an AIMD system.

A.5 Cluster analysis

We extract lithium-ion-centered clusters from MD simulation boxes using spatial criteria. Starting from an initial Li⁺ center, an initial cluster is formed by the inclusion of neighboring units of EC, PF₆⁻, or other Li⁺ ions based on a 4.5 Å cutoff between the lithium center and the respective center of mass of the unit (using simply the coordinates in the case of another lithium ion). The periodic boundary condition of the simulation box is implemented by applying the minimum image convention while calculating distances among different units. For the included EC units, a 2.5 Å bond distance cutoff is verified between the Li⁺ center and at least one oxygen atom of the EC molecule. EC molecules, that are part of the initial cluster but fail the bond distance test, are eliminated. The same filtering is applied for included PF₆⁻ units in the initial cluster using a 2.5 Å bond distance cutoff between the Li⁺ center and at least one of the fluorine atoms in the PF₆⁻ unit.

Once the initial cluster seed is obtained, further contiguity of the included Li⁺ or EC units are checked. The cluster can, therefore, expand using the Li⁺ or the EC units as new centers as long as other neighboring units of EC, PF₆⁻, or Li⁺ satisfy the 4.5 Å cutoff between the new center and the respective neighboring unit's center of mass. The bond distance test step is executed again. This cluster expansion is applied until no new neighboring Li⁺ or EC unit qualifies as a new center. This clustering algorithm ensures that the same final cluster is obtained irrespective of the sequence in the clustering process. Once a cluster is finalized and extracted, the included atoms are removed from the box to prevent double counting. The next available Li⁺ is now considered for the next cluster and so on. In this clustering procedure, unlike Li⁺ or EC, the PF₆⁻ units do not qualify as a subsequent clustering center to avoid aggregates of multiple cations and anions as part of the clusters.

Resulting statistics for the composition of Li-centered clusters from QRNN and OPLS4 MD are shown in Figures 1(d), 10, and 11, highlighting the differences between the two MD trajectories and the trends emerging from increasing LiPF₆ concentration from neat EC to 2 M LiPF₆ in EC. Beyond the observation of increased abundance of multi-Li complexes with increasing concentration, the difference in coordination environments of Li⁺ between QRNN and OPLS4 is evident from the specificity in numbers of EC molecules complexed with Li⁺ in Li₁ or Li₂ clusters. This is qualitatively different in OPLS4 where a diverse range of coordination environments is observed.

Fig. 12 shows a bar graph of the relative abundance of the most prominent clusters from the QRNN trajectory at varying LiPF₆ concentrations in EC. The most prominent composition across all concentrations is LiEC₄, comprising more than 80% of Li clusters at 0.5 M LiPF₆. As the concentration of LiPF₆ is increased, a variety of new compositions occur in the MD trajectory. Among the concentrations considered, the maximum variety of compositions is observed at 2

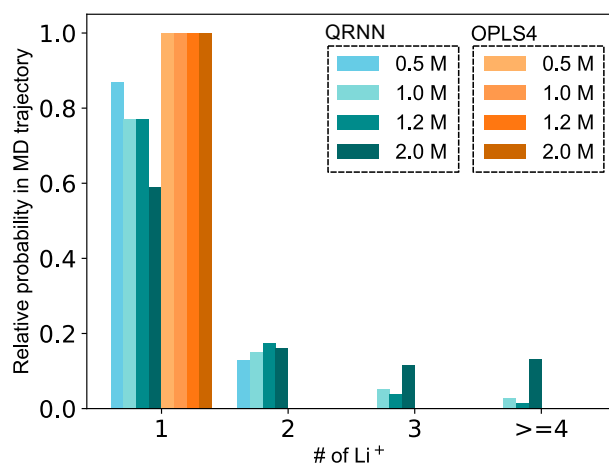


Figure 10: Relative abundance of clusters with different numbers of Li⁺ as fractions of the total number of clusters in QRNN or OPLS4 trajectories at a given concentrations of LiPF₆ in EC. The statistics shown here are complementary to Fig. 1(d) where the % distribution of all Li⁺ among different Li_n cluster types is shown.

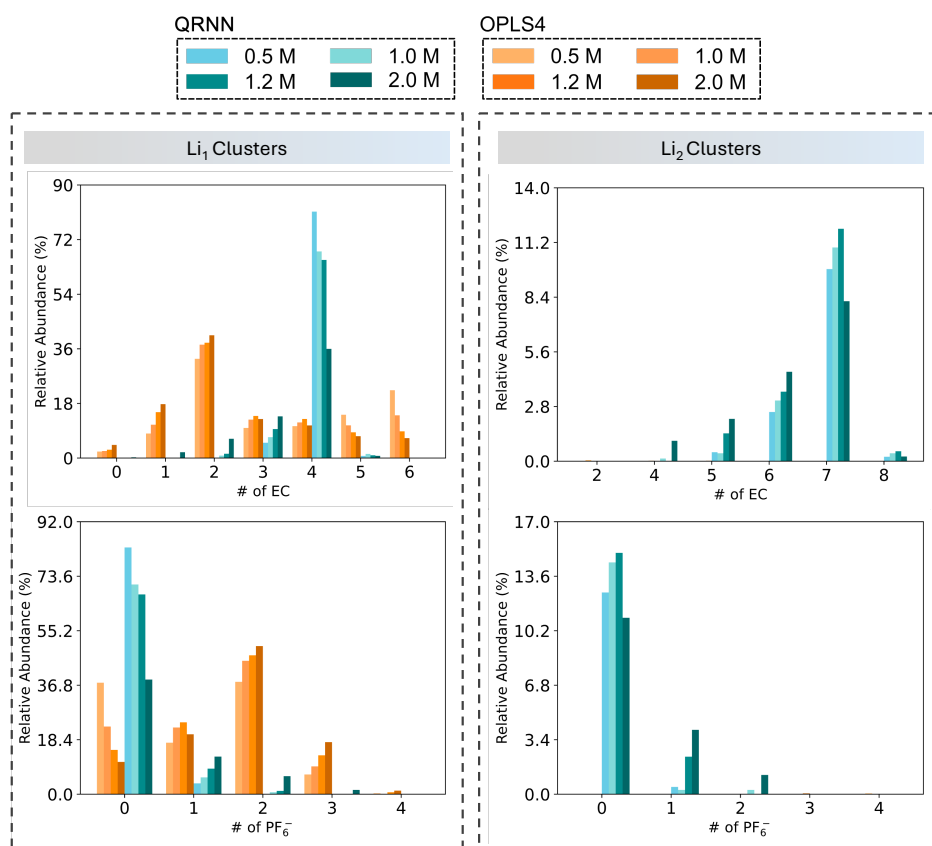


Figure 11: Relative abundance of clusters with specific numbers of EC and PF₆⁻ as fractions of the total number of clusters in QRNN or OPLS4 trajectories at a given concentrations of LiPF₆ in EC. The distribution of numbers of EC and PF₆⁻ are shown separately for Li₁ and Li₂ clusters. Compared to OPLS4, QRNN clusters have more defined coordination environments of Li⁺, manifested by the sharp distribution in number of EC and PF₆⁻. OPLS4 shows an excessive preference for contact ion pairs of Li⁺ and PF₆⁻.

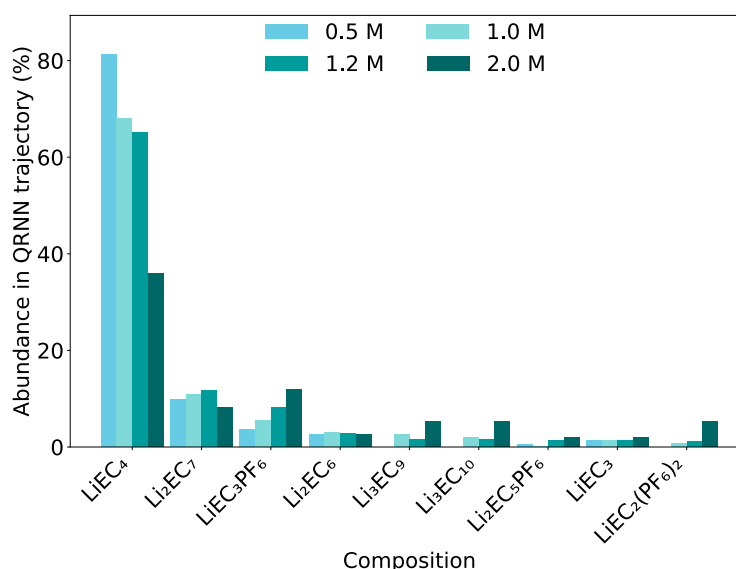


Figure 12: Relative abundance of the dominant cluster compositions as fractions of the total number of clusters from the QRNN trajectories at different concentrations of LiPF₆ in EC. Increase in the concentration leads to enhanced compositional diversity.

M, although the defined coordination numbers of Li⁺ (as discussed earlier) are still maintained. Among the multi-Li complexes, Li₂EC₇ is the most abundant followed by Li₂EC₆.

A.6 NMR details and data

A.6.1 Previous Li⁺ solvation studies by NMR

Lithium-ion solvation governs electrolyte stability, ionic transport, and ultimately the safety and lifetime of rechargeable batteries. Nuclear magnetic resonance (NMR) spectroscopy is uniquely suited to investigate this phenomenon, as it probes both static coordination motifs around Li⁺ and dynamic solvent-ion exchange processes that modulate conductivity. Solution-state NMR captures ensemble-averaged shifts and diffusion coefficients that delineate preferential binding of ethylene carbonate (EC) to Li⁺ over linear carbonates and other co-solvents^{81,82}, while relaxation analyses quantify exchange rates on timescales ranging from nanoseconds to seconds⁸³. Previous studies by Bogle et al.⁸⁴ and Yang et al.⁸² investigated Li⁺ solvation using solution-state NMR, monitoring chemical shift changes in ¹⁷O⁸⁴, ¹H, and ¹³C⁸² respectively. The authors observed gradual shifts in resonance positions rather than peak splitting indicating that the local environments around Li⁺ undergo rapid dynamic exchange at room temperature. This exchange leads to chemical shift averaging and an inability to resolve distinct solvation motifs. To overcome this limitation and capture the molecular complexity of lithium-ion coordination, we employed low-temperature frozen-state ¹³C solid-state NMR (ssNMR), enabling direct observation of different coordination environments by arresting solvent exchange processes.

A.6.2 Additional NMR data

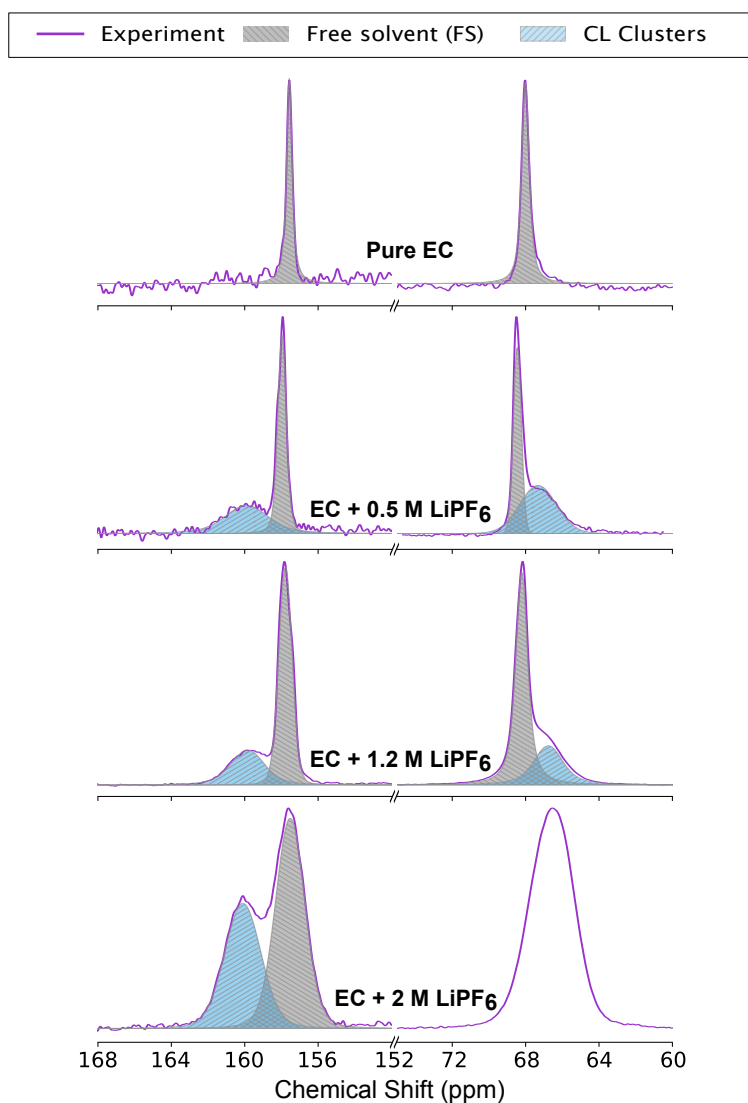
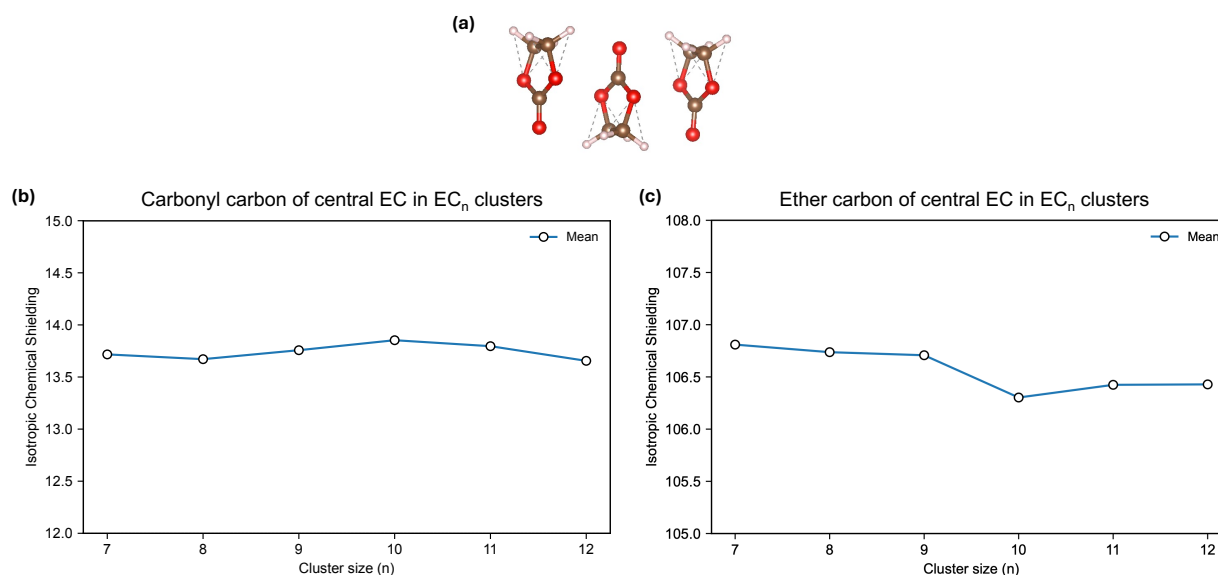


Figure 13: Deconvolution of the experimental ¹³C NMR spectra into free solvent and CL cluster components for both carbonyl and ether peaks at different LiPF₆ concentrations in EC where the peaks are fitted to Voigt profiles (except for at 2 M concentration where we observe a single broad ether peak).

Sample	¹³ C site	¹³ C shift (ppm)	FWHM (Hz)
0.5 M LiPF ₆	Uncoordinated C=O	158.0	80
	Li-coordinated C=O	159.8	482
	Uncoordinated C–O	68.5	80
	Li-coordinated C–O	67.3	390
1.2 M LiPF ₆	Uncoordinated C=O	157.8	144
	Li-coordinated C=O	160.0	300
	Uncoordinated C–O	68.3	80
	Li-coordinated C–O	67.5	400
2.0 M LiPF ₆	Uncoordinated C=O	157.7	280
	Li-coordinated C=O	160.0	325
	Uncoordinated C–O	N/A—single peak	—
	Li-coordinated C–O	66.6	390

Table 2: ¹³C chemical shifts and linewidths for different concentrations of LiPF₆ in EC.Figure 14: (a) QRNN Optimized antiparallel configuration of EC₃. (b-c) Isotropic chemical shielding (raw numbers) of the central EC molecule in clusters of configuration EC_n for the (b) carbonyl carbon and the (c) ether carbon.

A.6.3 NMR simulation with QRNN using solvent model based on physical arguments

The experimental NMR spectra at 0.5 M and 1.2 M concentrations of LiPF₆ in EC show that the solvent peak is well defined, represented by sharp and consistent peaks for both carbonyl and ether. This indicates that the free solvent has a consistent configuration at the quenched state, likely a small motif where the EC molecules still experience other solvent molecules in their environment. Therefore, an isolated EC molecule is an incomplete description of the free solvent EC molecules. During the experimental quenching process, despite the short quenching time, the EC molecules may go through local rearrangements involving their nearest neighbors. The smallest configuration where an EC molecule experiences the effect of the environment is EC₃. The most stable EC₃ configuration upon QRNN optimization with a tight optimization threshold is shown in Fig. 14(a). Similar head-to-tail motif has been experimentally verified in the crystal structure of EC⁸⁵ and would be reasonable in amorphous solid EC at quenched temperatures. In 2 M salt concentration, however, the solvent peak drastically broadens, indicating the lack of a consistent ordered motif repeating throughout the solvent.

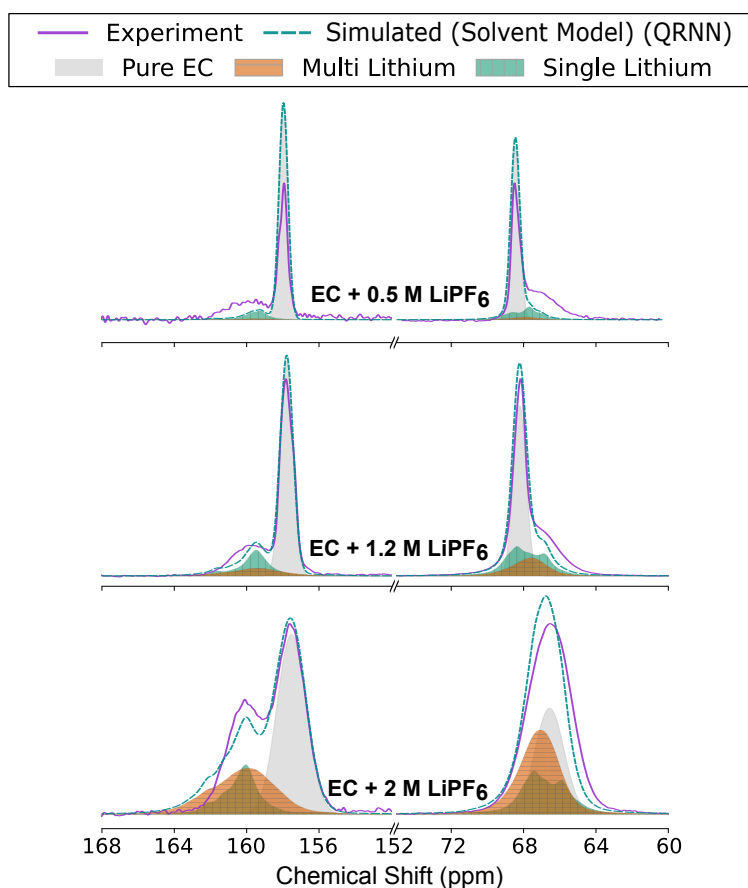


Figure 15: Simulated NMR results for 0.5 M, 1.2 M, and 2 M LiPF₆ in EC solvent using QRNN and modeling the free solvent.

For 0.5 M and 1.2 M, we model the free solvent using the optimized EC₃ configuration, where each solvent EC molecule's chemical shielding environment is represented by the isotropic shielding of the central EC molecule in EC₃.

We further verify the robustness and convergence of the EC environment as far as chemical shielding is concerned. Larger free EC clusters such as EC₆, EC₉, EC₁₂, and EC₁₅ are constructed using EC₃ as building blocks. We find that the chemical shielding for the carbon atoms in the central EC molecules is consistent, given that the antiparallel EC₃ structure is intact. For 2 M concentration, due to the lack of a sharp peak for both ether and carbonyl, we model the free solvent as the central EC in extracted solvent EC clusters from MD with size ranging from EC₉-EC₁₂. These are the majority free solvent cluster configurations as clusters are extracted with a cutoff radius 6.5 Å around the central EC (consistent with EC-EC RDF in the QRNN trajectory). We take the average chemical shielding of the central EC molecule across 120 EC_n clusters that are QRNN optimized. The consistency of the average chemical shielding across EC_n configurations is shown in Fig 14(b-c).

Following identical protocol to the NMR results shown in Fig. 3 for the CL clusters, and augmenting the solvent peak with aforementioned structural units we calculate the NMR spectra for all experimental concentrations of LiPF₆. The simulated spectra along with the different components contributing to the spectra are shown in Fig. 15. The obtained spectra accurately describe the experiments.

A.7 Electrochemistry benchmarking and results

A.7.1 Gas phase benchmarking of DFT methodologies versus DLPNO-CCSD(T) for reduction potentials

The gas-phase electronic energy contributions to the reduction potentials of selected Li-ion complexes are utilized to validate DFT methods and characterize complexes observed during MD. Herein, we consider just the gas phase electronic energy component of the reduction potentials denoted ΔU_{red} , computed as:

$$\Delta U_{red} = U_{red} - U_{ox} \quad (2)$$

where U_{red} and U_{ox} are the computed internal (electronic) energies of the reduced and oxidized complexes, respectively. The errors in the DFT ΔU_{red} values for a series of optimized Li-EC clusters versus DLPNO-CCSD(T) are reported in Table 3. From these errors, we observe that the best-performing DFT methodologies for redox potentials are many of the same best-performing methods for association energies of the same clusters. Namely, ω B97X-D3BJ—the method on which QRNN was ultimately trained—continues to perform well for redox properties, as do B2PLYP+D3 and CAM-B3LYP+D3.

Complex	ΔU_{red}	$\Delta \Delta U_{red}$ vs DLPNO-CCSD(T), TZ					
	DLPNO TZ	B2PLYP+D3	M06-2X+D3	SCAN+D3	CAM-B3LYP+D3	ω B97X-D3BJ	PBE0+D3
LiEC ₁	-84.78	-2.03	-0.35	-2.74	-3.54	-3.49	-4.51
LiEC ₂	-55.86	-1.48	-0.32	-4.55	-1.9	-1.67	-4.3
LiEC ₃	-48.56	-0.66	-10.48	-6.99	-0.73	4.7	-4.1
LiEC ₄	-62.14	1.14	-2.9	-2.42	-1.98	0.90	1.52

Table 3: Gas-phase electronic energy contributions to the reduction potentials for several Li-EC coordination complexes from DLPNO-CCSD(T)/aug-cc-pVTZ and the absolute errors versus these values for various DFT methods (def2-TVZPD basis). All values in kcal/mol; reductions are from the +1 to neutral states.

A.7.2 Benchmarking of electronic structure and implicit solvation methods for Li-EC complex reduction potentials

To further assess the reliability of ω B97X-D3BJ/def2-TZVPD and different implicit solvation methods for the reduction potentials of Li-ion complexes with EC ligands, we compared these values against the corresponding values using DLPNO-CCSD(T)/aug-cc-pVTZ. The reduction potentials in Table 4, which for convenience neglect zero-point and entropic contributions to the component species' free energies, contrast these two electronic structure methods as well as two methods for implicit solvation: C-PCM and SMD. Both continuum solvent methods employ a static dielectric constant of 33.6, consistent with the conditions of the experimental reference FeCp^{+/0} couple. Additional parameters for SMD were taken to be those from a previous parameterization for EC⁵⁹.

		DLPNO/aug-cc-pVTZ		ω B97X-D3BJ/def2-TZVPD	
		C-PCM	SMD	C-PCM	SMD
N_{Li}	N_{EC}	E (V vs. Li/Li ⁺)	E (V vs. Li/Li ⁺)	E (V vs. Li/Li ⁺)	E (V vs. Li/Li ⁺)
1	1	0.34	0.57	0.58	0.55
1	2	0.45	0.52	0.43	0.51
1	3	0.17	0.35	0.12	0.30
1	4	-0.04	0.15	0.04	0.26
2	4	0.54	0.61	0.56	0.63
2	5	0.52	0.65	0.52	0.65

Table 4: One-electron reduction potentials vs. Li/Li⁺ for select Li-EC complexes, comparing DLPNO-CCSD(T)/aug-cc-pVTZ and ω B97X-D3BJ/def2-TZVPD methods with C-PCM and SMD solvation.

From Table 4, it is evident that the reduction potentials yielded by ω B97X-D3BJ calculations are in overall good agreement with those yielded by DLPNO-CCSD(T), when controlling for implicit solvation method. The lone exception to this is LiEC₁ with DLPNO and C-PCM; however, this is not a chemical species that was statistically relevant in bulk solution from the QRNN MD simulations. Moreover, we largely see strong consistency among the reduction potentials predicted using the SMD solvation scheme across the two electronic structure methods. For this reason, coupled with the relative computational efficiency of the DFT approach as compared to DLPNO-CCSD(T), we report all reduction potentials hereafter using the ω B97X-D3BJ/def2-TZVPD/SMD combination of methodologies with confidence in its agreement with benchmark-level values.

A.7.3 Referenced reduction potentials of Li/EC coordination complexes

The one-electron reduction potentials of optimized Li/EC complexes using ω B97X-D3BJ/def2-TZVPD/SMD are shown in Table 5. The E_1 value corresponds to the one-electron reduction of the species listed and serve as the data on which Fig. 4 is based whereas the E_2 value is the potential associated with a second one-electron reduction to a two-electron reduced species. E_2 values were only obtained for a small subset of the clusters for which there was no spontaneous EC ligand dissociation or ring-opening.

Complex	E_1 (V vs. Li/Li ⁺)	E_2 (V vs. Li/Li ⁺)
LiEC1	0.65	
LiEC2	0.63	
LiEC3	0.37	
LiEC4	0.25	
LiEC3PF ₆	0.33	
Li ₂ EC ₄	0.72	0.95
Li ₂ EC ₅	0.68	2.14
Li ₂ EC ₆	0.77	2.97
Li ₂ EC ₇	0.71	
Li ₃ EC ₉	0.89	1.17
Li ₃ EC ₁₀	0.81	

Table 5: Calculated one-electron reduction potentials referenced to Li/Li⁺ of Li-EC coordination complexes using ω B97X-D3BJ/def2-TZVPD with SMD ($\epsilon = 33.6$). Each unreduced (oxidized) species has net charge $n+$, where n is the number of Li-ions.

These reduction potentials are broadly analyzed in the main text. Additionally, while the Li₁EC_{1–2} complexes demonstrate fairly positive reduction potentials, they are unlikely to be relevant to SEI formation given the very low statistical probability of forming in electrolyte solutions. We additionally computed the reduction potential for the most statistically relevant PF₆-containing complex, LiEC₃PF₆, to be comparable to those of Li₁EC_{3–4}. While this reduction potential is still not competitive with those computed for Li₂₊ complexes, it is notable that PF₆ anions may not be prohibitive of reductive decomposition reactions.

We note here that the referenced reduction potentials presented in this work differ from other estimates in the literature. For example, the work of Hou et al.⁷⁰ estimates the LiEC₄ reduction potential to be 0.49 V vs. Li/Li⁺. The ca. 0.25 V difference from our calculations is well within the variance expected given the different DFT functional (B3LYP), basis set (6-31+G(d)), solvation method (PCM), and dielectric constant (90). Moreover, we anticipate that any reasonable choice of DFT protocol for computing reduction potentials would yield the same qualitative conclusions regarding the more positive potentials associated with multi-Li complexes as compared to Li₁ complexes.

The singly reduced Li₂EC_{4–6} and Li₃EC₉ species all demonstrated an even more favorable second one-electron reduction process. The doubly reduced Li₂EC₇ complex exhibited spontaneous dissociation of an EC ligand and its second reduction potential is therefore not included in this work although it merits further study given the statistical prominence of this species. Upon geometry optimization of the doubly reduced (net neutral) Li₂EC_{5–6} species, EC was spontaneously activated, including ring opening transformations. Given the favorable reduction potentials associated with these concerted reduction-EC activations, this merits additional mechanistic studies that are beyond the scope of the current work.

A.8 Future work

In this work, we have provided compelling theoretical and experimental evidence that a significant concentration of multi-lithium clusters is present in electrolytes composed of a mixture of ethylene carbonate solvent and LiPF₆ salt. We have carried out an initial exploration of the electrochemistry of the mixture and shown that the reduction potential of these clusters are significantly more accessible than that of the single lithium species, and hence such reactions are likely to play an important role in SEI formation. Further work is ongoing to elaborate the relevant electrochemical reactions, and ultimately to perform MLFF based simulations of the electrolyte in actual battery environment, with the incorporation of electron transfer dynamics into the simulations.

A broad range of electrolyte compositions are used in batteries, of which we have only addressed a few in the present study. Alternative lithium salts such as LiFSI, LiTFSI, and LiBF₄ might, for example, show different ratios of multi-Li⁺

complexes to CIPs. Li⁺ complexation statistics will also likely be sensitive to the presence of additives such as VC and FEC. The sensitivity of the complexation to changes in electrolyte composition also depends on the quantum mechanical details of the interactions which hold the Li⁺ clusters together, a topic deserving of further study.

The impact of Li⁺ clustering where it was not expected has consequences for the practical performance of cells, both beneficial and detrimental, depending on the cell context.

Lithium ion clustering, particularly in highly concentrated electrolytes (HCEs) and densely packed ion-cluster electrolytes (DPIEs), has been cited to offer several advantages for battery performance by influencing interfacial properties and charge transport mechanisms. For instance, DPIEs are designed to promote fast Li⁺ desolvation and the formation of robust electrode-electrolyte interphases, while also suppressing polysulfide dissolution in lithium-sulfur batteries⁸⁶. In solvent-in-salt electrolytes, the structure resulting from anions participating in cation solvation leads to preferential anion reduction to form the solid-electrolyte interphase (SEI) and increased oxidative stability at the cathode⁸⁷. In low-polarity ether-based electrolytes, aggregated clusters of Li⁺ and FSI⁻ anions can accumulate at the cathode surface and work to shield free solvent molecules from direct electrode contact. This kinetically favors the decomposition of anions over solvents and is hypothesized to enhance high-voltage stability⁸⁸. This clustered structure is also hypothesized to facilitate structural/hopping lithium transport in HCEs, where lithium-anion-lithium coordination networks are formed due to the scarcity of free solvent molecules, which can lead to higher lithium transference numbers (t^+)^{87,89,90}. It remains to be seen what influence, if any, multi-Li complexation has on the overall transference number.

However, lithium-ion clustering can also present drawbacks for battery performance. While beneficial for interfaces, highly concentrated electrolytes often exhibit lower ionic conductivity and significantly higher viscosity than traditional dilute electrolytes^{87,90}. This increased clustering effectively reduces the number of mobile charge carriers. It increases the hydrodynamic radius of ions, thereby hindering their overall transport and limiting fast-charging capabilities and power densities^{87,91}. Furthermore, in localized high concentration electrolytes (LHCEs), the addition of diluents can significantly decrease the Li⁺ transference number by disrupting the concerted cation-hopping mechanism^{87,92}. In some polyelectrolyte solutions, strong ion clustering can lead to a “negative lithium transference number”, where the net lithium motion is observed to be opposite to the electric field due to lithium ions being “dragged” by negatively charged clusters⁹³. At high current densities, electrolytes with more significant clustering tend to result in thicker and more porous lithium deposition morphologies, which can compromise cycling stability and lead to increased loss of lithium inventory⁹¹. This is linked to the formation of sharp concentration gradients at the electrode-electrolyte interface, promoting heterogeneous current density distributions⁹¹.

These previous efforts describe concentration-driven clustering in these “extreme” examples. The current work outlined herein indicates it may be a far more common behavior than previously understood, and the impact of this “subtle clustering” needs far more research. Simulations based on a more complete representation of the battery environment, and including electron transfer and reactive chemistry, will enable the impact of multi-lithium clusters on both ion dynamics in the electrolyte and formation of the SEI to be elucidated.

It may be the goal of future solvent engineers to moderate the formation of these complexes, or to use solvent blends that stabilize mono-lithium and keep the electrolyte as similar as possible to the previous picture which does not include Li⁺ clustering. However, the existence of hitherto-unknown Li⁺ clusters in our battery electrolytes would demonstrate that, even in a chemical system that we carry in our pockets, we still have much more to learn.

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